

Exp No: 5 INTEGRATION WITH EXTERNAL SYSTEMS USING HYPERLEDGER FABRIC

AIM

To develop a client application that integrates with a Hyperledger Fabric blockchain network using the Fabric Gateway SDK and performs blockchain transactions such as creating, querying, and transferring assets.

SOFTWARE REQUIREMENTS

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js Version 20 or later
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Fabric Gateway SDK (Node.js)
- Visual Studio Code

PROCEDURE

Step 1: Upgrade Node.js to Version 20+

Install Node.js version 20 or later using NodeSource.

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
node -v
Ensure that the installed version is v20.x.x or later before continuing.
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ node -v
v20.20.2
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 2: Start the Fabric Network

Navigate to the Hyperledger Fabric test network, stop any previously running network, and start the network with the application channel and Certificate Authority.

```
cd ~/fabric-dev/fabric-samples/test-network
./network.sh down
```

```
./network.sh up createChannel -ca
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh
up createChannel -ca
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI de
lay of '3' seconds and using database 'leveldb with crypto from 'Certificate
Authorities'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
CA_LOCAL_VERSION=v1.5.17
CA_DOCKER_IMAGE_VERSION=v1.5.17
Generating certificates using Fabric CA
[+] up 4/4
  ✓ Network fabric_test Created 0.0s
  ✓ Container ca_org2 Started 0.3s
  ✓ Container ca_orderer Started 0.4s
  ✓ Container ca_org1 Started 0.3s
+ fabric-ca-client getcainfo -u https://admin:adminpw@localhost:7054 --caname
ca-org1 --tls.certfiles /home/naveen/fabric-dev/fabric-samples/test-network/
organizations/fabric-ca/org1/ca-cert.pem
2026/09/01 17:24:09 [INFO] Configuration file location: /home/naveen/fabric-d
ev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.c
om/fabric-ca-client-config.yaml
2026/09/01 17:24:09 [INFO] TLS Enabled
2026/09/01 17:24:09 [INFO] Stored root CA certificate at /home/naveen/fabric-d
ev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.
com/msp/cacerts/localhost-7054-ca-org1.pem
2026/09/01 17:24:09 [INFO] Stored Issuer public key at /home/naveen/fabric-de
v/fabric-samples/test-network/organizations/peerOrganizations/org1.example.co
m/msp/IssuerPublicKey
2026/09/01 17:24:09 [INFO] Stored Issuer revocation public key at /home/navee
n/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1
.example.com/msp/IssuerRevocationPublicKey
+ res=0
Creating Org1 Identities
Enrolling the CA admin
+ fabric-ca-client enroll -u https://admin:adminpw@localhost:7054 --caname ca
-org1 --tls.certfiles /home/naveen/fabric-dev/fabric-samples/test-network/org
anizations/fabric-ca/org1/ca-cert.pem
2026/09/01 17:24:09 [INFO] Created a default configuration file at /home/nave
en/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org
1.example.com/fabric-ca-client-config.yaml
```

Step 3: Deploy the JavaScript Chaincode

Deploy the Asset Transfer Basic JavaScript chaincode.

Name:Naveen CH

Roll No:231901033

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-  
basic/chaincode-javascript
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh  
deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-j  
avascript  
Using docker and docker-compose  
deploying chaincode on channel 'mychannel'  
executing with the following  
- CHANNEL_NAME: mychannel  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
- CC_SEQUENCE: auto  
- CC_END_POLICY: NA  
- CC_COLL_CONFIG: NA  
- CC_INIT_FCN: NA  
- DELAY: 3  
- MAX_RETRY: 5  
- VERBOSE: false  
executing with the following  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
+ '[' false = true ']'  
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basi  
c/chaincode-javascript --lang node --label basic_1.0  
+ res=0  
Chaincode is packaged  
Installing chaincode on peer0.org1...  
Using organization 1  
+ peer lifecycle chaincode queryinstalled --output json
```

Step 4: Navigate to the Application Directory

Move to the JavaScript application directory, list the files, and inspect package.json.

```
cd ../asset-transfer-basic/application-gateway-javascript  
ls
```

```
cat package.json
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ../asset-
transfer-basic/application-gateway-javascript
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/appli
cation-gateway-javascript$ ls
eslint.config.mjs package.json src
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/appli
cation-gateway-javascript$ cat package.json
{
  "name": "asset-transfer-basic",
  "version": "1.0.0",
  "description": "Asset Transfer Basic Application implemented in JavaScrip
t using fabric-gateway",
  "engines": {
    "node": ">=20"
  },
  "scripts": {
    "lint": "eslint src",
    "pretest": "npm run lint",
    "start": "node src/app.js"
  },
  "engineStrict": true,
  "author": "Hyperledger",
  "license": "Apache-2.0",
  "dependencies": {
    "@grpc/grpc-js": "^1.14.0",
    "@hyperledger/fabric-gateway": "^1.10.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.5.0",
    "eslint": "^9.5.0",
    "globals": "^15.6.0"
  }
}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/appli
cation-gateway-javascript$
```

Step 5: Clean Install Node.js Dependencies

Remove the existing node_modules directory and package lock file, then install the required Node.js packages.

```
rm -rf node_modules package-lock.json
npm install
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ rm -rf node_modules package-lock.json
npm install
npm warn deprecated eslint@9.39.5: This version is no longer supported. Please see https://eslint.org/version-support for other options.

added 123 packages, and audited 124 packages in 7s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
```

Step 6: Run the Client Application

Run the application using the available start script. If a start script is not available, run the application directly using its entry point.

```
npm start
node src/app.js
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ ~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript
-bash: /home/naveen/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript: No such file or directory
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ npm start

> asset-transfer-basic@1.0.0 start
> node src/app.js

channelName:      mychannel
chaincodeName:    basic
mspId:            Org1MSP
cryptoPath:       /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com
keyDirectoryPath: /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com
```



```

naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ node src/app.js
channelName:      mychannel
chaincodeName:    basic
mspId:            Org1MSP
cryptoPath:       /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com
keyDirectoryPath: /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/User1@org1.example.com/msp/keystore
certDirectoryPath: /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/User1@org1.example.com/msp/signcerts
tlsCertPath:      /home/naveen/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
peerEndpoint:     localhost:7051
peerHostAlias:    peer0.org1.example.com

--> Submit Transaction: InitLedger, function creates the initial set of assets on the ledger
*** Transaction committed successfully

```

Step 7: Initialize and Query the Ledger

The client application invokes the InitLedger transaction to initialize the ledger with sample assets. It then evaluates the GetAllAssets transaction to retrieve the assets stored in the world state.

```

naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ cd ~/fabric-dev/fabric-samples/test-network
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$

```

Step 8: Create a New Asset

The client application submits a CreateAsset transaction to the blockchain network. The transaction creates a new asset in the ledger with the required asset details.

```

naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
2026-09-01 17:33:12.073 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"{\n  \"ID\": \"asset20\", \n  \"Color\": \"Green\", \n  \"Size\": 8, \n  \"Owner\": \"Arun\", \n  \"AppraisedValue\": 1500}"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$

```

Step 9: Transfer Asset Ownership

The application submits a TransferAsset transaction to transfer ownership of an existing asset to another owner. The transaction updates the asset information stored in the world state.

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"TransferAsset","Args":["asset20","Naveen"]}'
Error: endorsement failure during invoke. response: status:500 message:"The asset asset20 does not exist"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

Step 10: Query the Updated Asset

The client application evaluates the ReadAsset transaction to retrieve the updated asset information. The returned information is displayed by the application to verify that the ownership change was successfully recorded on the blockchain.

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset20"]}'
Error: endorsement failure during query. response: status:500 message:"The asset asset20 does not exist"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

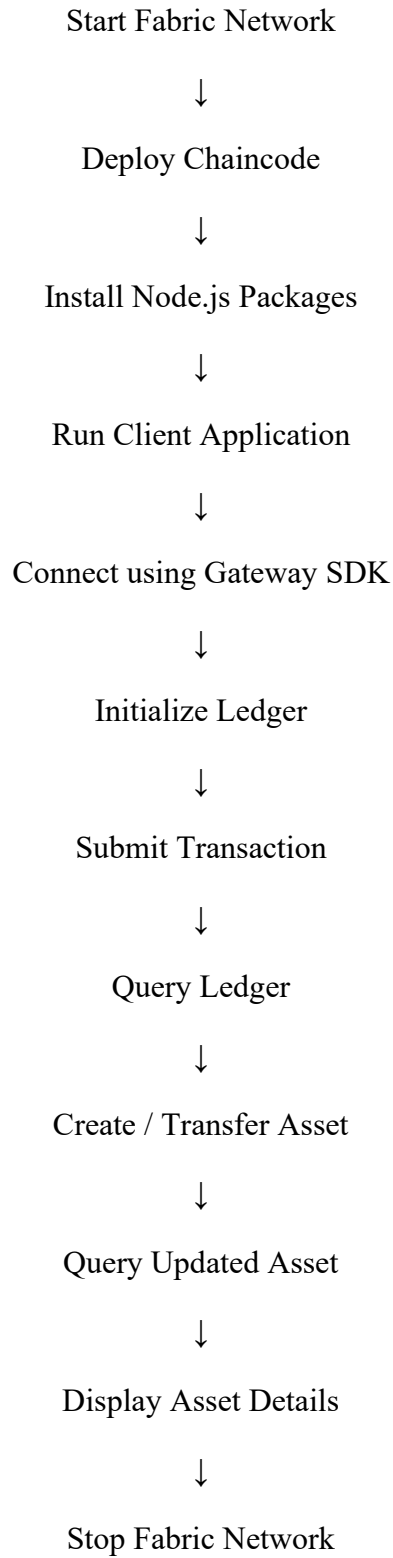
Step 11: Stop the Fabric Network

Navigate back to the Fabric test network directory and stop and remove the Fabric network.

```
cd ../../test-network
./network.sh down
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset20"]}'
Error: endorsement failure during query. response: status:500 message:"The asset asset20 does not exist"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 10/10
✓Container peer0.org1.example.com      Removed      1.0s
✓Container orderer.example.com         Removed      0.5s
✓Container ca_org2                     Removed      3.4s
✓Container ca_orderer                  Removed      3.6s
✓Container ca_org1                     Removed      3.7s
✓Container peer0.org2.example.com      Removed      0.9s
✓Volume compose_peer0.org1.example.com Removed      0.0s
✓Volume compose_peer0.org2.example.com Removed      0.0s
✓Network fabric_test                   Removed      0.3s
✓Volume compose_orderer.example.com    Removed      0.0s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
```


APPLICATION WORKFLOW



RESULT

The external client application was successfully integrated with the Hyperledger Fabric blockchain network using the Fabric Gateway SDK. The application established a secure connection, initialized and queried the ledger, created and transferred assets, and retrieved updated asset information, demonstrating how external applications can interact with a permissioned blockchain network.